

Formal Verification: FROST Threshold Schnorr Signatures

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Abstract

We formalize the FROST (Flexible Round-Optimized Schnorr Threshold) signature scheme used in the Lux FROST precompile at address `0x0200...000C`. We state correctness, unforgeability, robustness, and share refresh as axioms grounded in the discrete logarithm assumption in the random oracle model.

1 Introduction

FROST is the most gas-efficient classical threshold scheme on Lux, with 64-byte signatures and gas cost of $50,000 + 5,000$ per signer. It supports Bitcoin Taproot (BIP-340/341) compatibility and is used for cross-chain bridges and DAO governance.

2 Definitions

Definition 1 (ThresholdParams). *Parameters: t threshold, n total participants, with $t \leq n$ and $0 < t$.*

Definition 2 (SchnorrSig). *A Schnorr signature: nonce commitment R (32 bytes) and response scalar s (32 bytes).*

3 Theorems and Axioms

Axiom 3 (frost_correctness). *If t honest participants execute the protocol, the resulting signature verifies under the aggregate public key.*

Axiom 4 (frost_unforgeability). *With fewer than t shares, no adversary can produce a valid signature (discrete log hardness in ROM).*

Theorem 5 (threshold_bounded). *$t \leq n$ by construction.*

Theorem 6 (robustness). *With n participants and at most $n - t$ failures, at least t honest participants remain to sign.*

Theorem 7 (majority_threshold_safe). *Setting $t = \lfloor n/2 \rfloor + 1$ satisfies both $t \leq n$ and $t > n/2$.*

Axiom 8 (share_refresh_correctness). *After share refresh, new shares still produce valid signatures under the same aggregate public key.*

4 Lean Source

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/-
  FROST Threshold Signature Correctness

  Models the FROST precompile (src/precompiles/frost/).

  FROST: Flexible Round-Optimized Schnorr Threshold signatures.
  t-of-n threshold: any t participants can produce a valid Schnorr signature,
  fewer than t cannot.

  Maps to:
  - src/precompiles/frost/contract.go: Verify function
  - src/precompiles/frost/IFROST.sol: Solidity interface
  - luxfi/threshold/protocols/frost: Keygen, Sign, Verify

  Properties:
  - Correctness: signature produced by t honest signers verifies
  - Unforgeability: t-1 signers cannot forge (in random oracle model)
  - Robustness: t honest signers always produce valid output
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Crypto.FROST

/-- Threshold parameters -/
structure ThresholdParams where
  t : Nat      -- threshold
  n : Nat      -- total participants
  ht : t ≤ n   -- threshold ≤ total
  ht0 : 0 < t -- need at least 1 signer

/-- Abstract Schnorr signature model -/
structure SchnorrSig where
  R : Nat      -- nonce commitment (32 bytes)
  s : Nat      -- response scalar (32 bytes)

/-- Verification (abstract Schnorr):  $s \cdot G = R + H(R, pk, msg) \cdot pk$  -/
axiom schnorr_verify : SchnorrSig → Nat → Nat → Bool

/-- Share: each participant holds a secret share -/
structure Share where
  index : Fin n
  value : Nat

/-- FROST correctness: if t honest participants execute the protocol,
the resulting signature verifies under the aggregate public key. -/
axiom frost_correctness :
  (p : ThresholdParams) (shares : List Share)
  (msg : Nat) (aggPK : Nat),
  shares.length ≤ p.t →
  -- All shares are valid (from DKG)
```

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    (sig : SchnorrSig), schnorr_verify sig aggPK msg = true

/-- FROST unforgeability: with fewer than t shares,
    cannot produce a valid signature (in ROM). -/
axiom frost_unforgeability :
  (p : ThresholdParams) (compromised : Nat),
  compromised < p.t →
  -- Cannot forge: fewer than t shares cannot produce a valid signature
  -- (discrete log hardness in random oracle model)
  (msg : Nat) (aggPK : Nat) (sig : SchnorrSig),
  schnorr_verify sig aggPK msg = false

/-- Threshold property: exactly t signers suffice -/
theorem threshold_exact (p : ThresholdParams)
  (signers : Nat) (h : signers ≤ p.t) :
  signers > 0 := by
  omega

/-- t-of-n: t must be at most n -/
theorem threshold_bounded (p : ThresholdParams) : p.t ≤ p.n := by
  omega

/-- Security margin: compromising t-1 is safe -/
theorem security_margin (p : ThresholdParams) (compromised : Nat)
  (h : compromised < p.t) : compromised + 1 ≤ p.n := by
  omega

/-- Robustness: if we have n participants and at most f < n-t+1 fail,
    we still have t honest participants to sign -/
theorem robustness (p : ThresholdParams) (failures : Nat)
  (h : failures ≤ p.n - p.t) :
  p.n - failures ≥ p.t := by
  omega

/-- Optimal threshold: t = ceil(n/2)+1 for majority threshold -/
theorem majority_threshold_safe (n : Nat) (hn : n > 0)
  (t : Nat) (ht : t = n / 2 + 1) :
  t ≤ n - t > n / 2 := by
  constructor <|> omega

/-- Key share refresh: after refresh, old shares are invalidated.
    This is a protocol property: the refreshed polynomial has the
    same constant term (same secret) but different shares. -/
axiom share_refresh_correctness :
  (p : ThresholdParams) (old_shares new_shares : List Share)
  (aggPK msg : Nat),
  new_shares.length = p.t →
  (sig : SchnorrSig), schnorr_verify sig aggPK msg = true

end Crypto.FROST

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Crypto/FROST.lean>
 White papers: <https://papers.lux.network>

Related white papers:

- `lux-universal-threshold-signatures.tex`